Fitness Business Success In Utah

Daniela Binns, Aubrey Coons, Sedona Dettwiler

Approximately 20% of new businesses fail within the first two years, 50% within five years, and roughly 65-70% by year 10

Main Research Question

Can neighborhood characteristics and business category predict whether a fitness business will thrive in Utah?

Thriving: currently open, has a Yelp rating ≥ 4.0, and has at least 10 reviews — a composite signal of quality and community engagement.

Data

- Yelp Fusion API — 1,081 Utah fitness businesses
  - Rating, review count, category, price tier, coordinates
  - 7 categories: gyms, yoga, pilates, crossfit, martial arts, rock climbing, personal trainers
- U.S. Census ACS 2022 (5-year) — ZCTA-level demographics
  - Median income, home value, education, age distribution, population

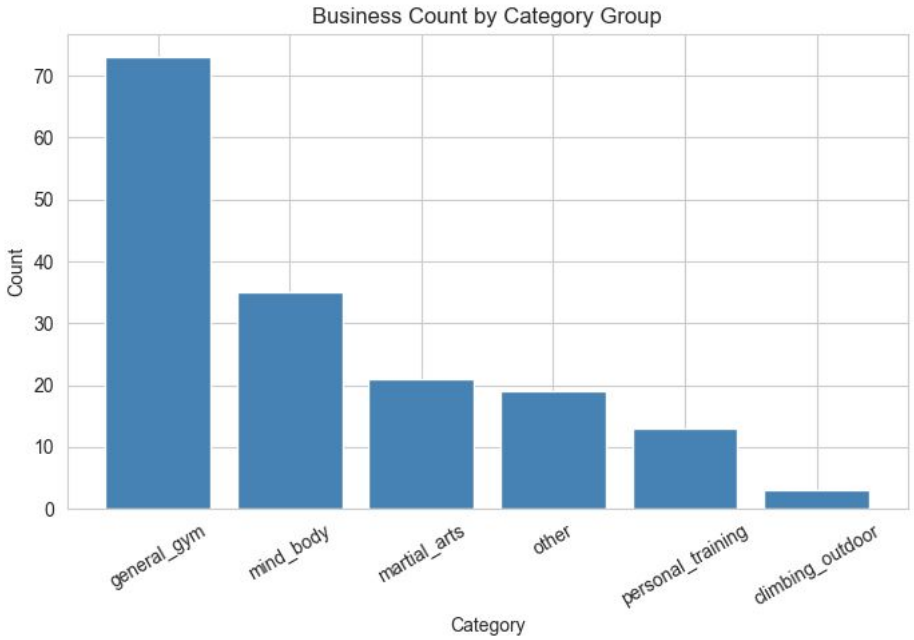
Derived features:

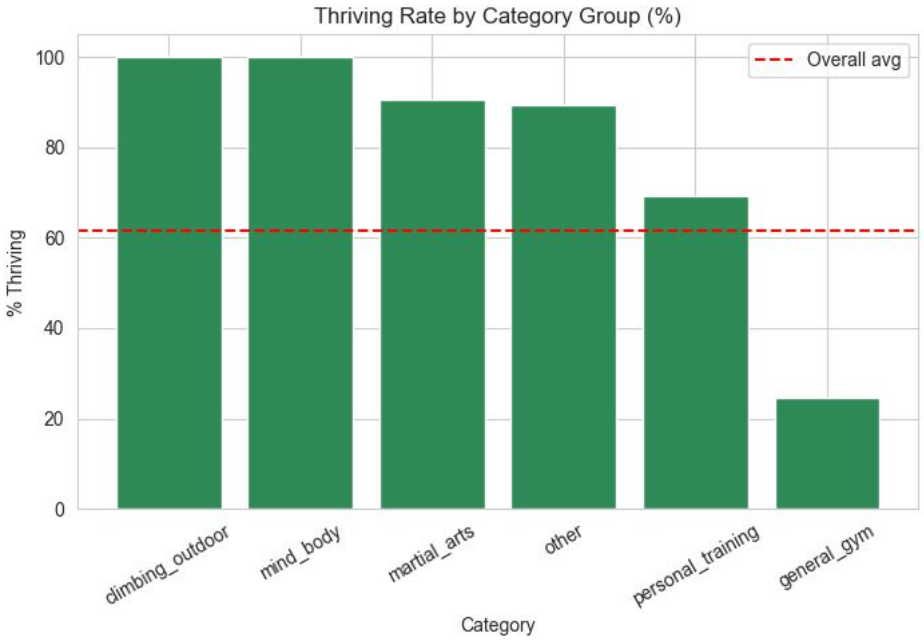
- Competition density (businesses within 1km and 3km haversine radius)
- market_gap = prime-age pop / (gyms in zip + 1)
- income_per_competitor = median income / (3km competitors + 1)
- gym_density_per_1k = gyms per 1,000 residents

Coverage: 84xxx Utah zip codes with population ≥ 5,000

Feature	Source	Description
`category_group`	Yelp	Grouped category: general_gym, mind_body, martial_arts, personal_training, climbing_outdoor, other
`price`	Yelp	Price tier (1–4, imputed with median)
`median_income`	Census ACS	Median household income by ZCTA
`median_home_value`	Census ACS	Median home value by ZCTA
`median_age`	Census ACS	Median age by ZCTA
`pct_bachelors`	Census ACS	% population with bachelor's degree
`pct_prime_gym_age`	Census ACS	% population aged 20–44
`total_pop`	Census ACS	Total population by ZCTA
`competition_1km`	Derived (haversine)	Fitness businesses within 1km radius
`competition_3km`	Derived (haversine)	Fitness businesses within 3km radius
`market_gap`	Derived	`pct_prime_gym_age × total_pop / (gyms_in_zip + 1)`
`gym_density_per_1k`	Derived	Gyms per 1,000 residents in zip
`income_per_competitor`	Derived	Median income / (competition_3km + 1)

EDA

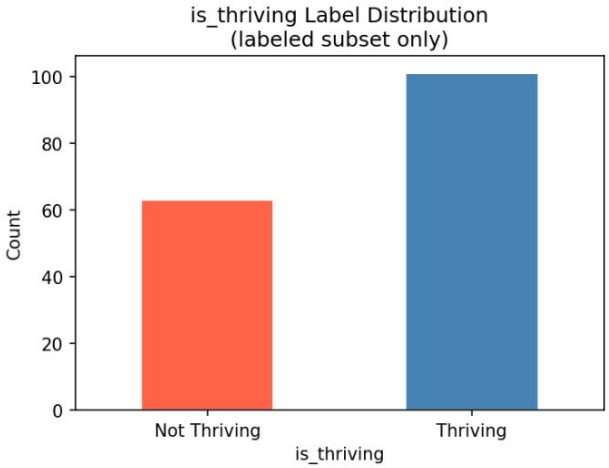

Business Count by Category Group
Count
0
10
20
30
40
50
60
70
general_gym
mind_body
martial_arts
other
personal_training
climbing_outdoor
Category

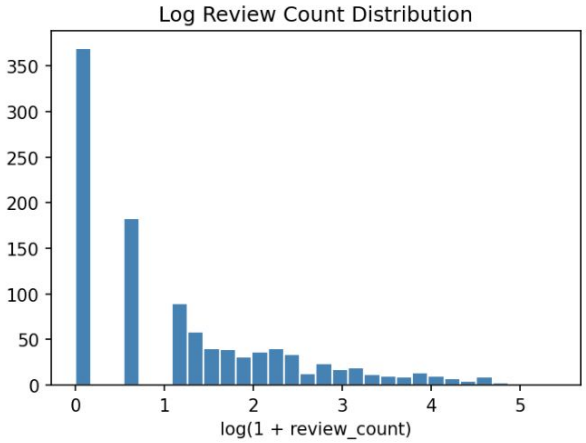

Thriving Rate by Category Group (%)
Overall avg
% Thriving
0
20
40
60
80
100
climbing_outdoor
mind_body
martial_arts
other
personal_training
general_gym
Category

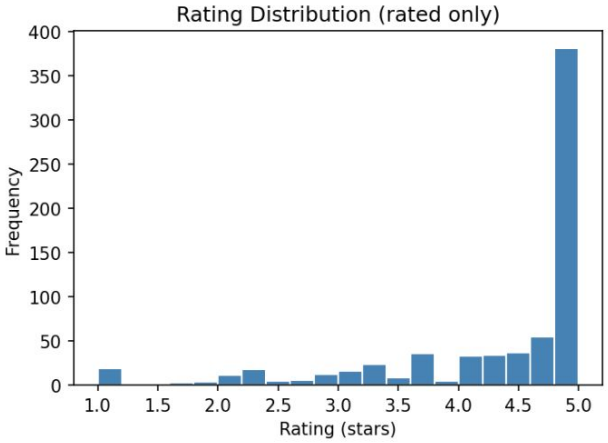
Correlation Matrix — All Numeric Features

	rating	review_count	is_closed	median_income	median_age	median_home_value	pct_bachelors	pct_prime_gym_age	competition_1km	competition_3km	competition_5km	market_gap	gym_density_per_1k	income_per_competitor	pop_per_competitor
rating	1.00	0.13		0.05	0.14	0.09	0.22	0.03	0.11	0.24	0.31	-0.10	0.08	-0.16	-0.14
review_count	0.13	1.00		-0.01	0.06	0.03	0.12	0.07	0.05	0.16	0.17	-0.05	0.05	-0.09	-0.07
is_closed															
median_income	0.05	-0.01		1.00	0.26	0.62	0.48	-0.58	0.00	-0.17	-0.15	-0.24	0.02	0.18	0.05
median_age	0.14	0.06		0.26	1.00	0.55	0.46	-0.52	0.10	0.01	0.02	-0.53	0.51	0.11	-0.16
median_home_value	0.09	0.03		0.62	0.55	1.00	0.75	-0.30	0.25	0.05	0.00	-0.33	0.59	0.05	-0.12
pct_bachelors	0.22	0.12		0.48	0.46	0.75	1.00	-0.01	0.47	0.47	0.45	-0.42	0.56	-0.10	-0.26
pct_prime_gym_age	0.03	0.07		-0.58	-0.52	-0.30	-0.01	1.00	0.28	0.49	0.46	0.42	0.09	-0.26	-0.08
competition_1km	0.11	0.05		0.00	0.10	0.25	0.47	0.28	1.00	0.64	0.49	-0.25	0.46	-0.35	-0.32
competition_3km	0.24	0.16		-0.17	0.01	0.05	0.47	0.49	0.64	1.00	0.89	-0.19	0.34	-0.46	-0.41
competition_5km	0.31	0.17		-0.15	0.02	0.00	0.45	0.46	0.49	0.89	1.00	-0.11	0.20	-0.42	-0.36
market_gap	-0.10	-0.05		-0.24	-0.53	-0.33	-0.42	0.42	-0.25	-0.19	-0.11	1.00	-0.50	0.03	0.27
gym_density_per_1k	0.08	0.05		0.02	0.51	0.59	0.56	0.09	0.46	0.34	0.20	-0.50	1.00	-0.08	-0.24
income_per_competitor	-0.16	-0.09		0.18	0.11	0.05	-0.10	-0.26	-0.35	-0.46	-0.42	0.03	-0.08	1.00	0.63
pop_per_competitor	-0.14	-0.07		0.05	-0.16	-0.12	-0.26	-0.08	-0.32	-0.41	-0.36	0.27	-0.24	0.63	1.00

1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4


is_thriving Label Distribution
(labeled subset only)
Count
0
20
40
60
80
100
Not Thriving
Thriving
is_thriving


Log Review Count Distribution
0
50
100
150
200
250
300
350
0
1
2
3
4
5
log(1 + review_count)


Rating Distribution (rated only)
Frequency
0
50
100
150
200
250
300
350
400
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Rating (stars)

Supervised Models

Model	Type	Key Hyperparameters Explored	Best Params	CV RMSE	Test RMSE	Test R^2
Ridge Regression	Regularized linear regression (baseline)	`alpha` : [0.1, 1.0, 10.0, 100.0]	`alpha = 10.0`	1.011	0.878	0.047
Random Forest	Ensemble tree-based regressor	`n_estimators` : [100, 200]; `max_depth` : [None, 10, 20]; `min_samples_split` : [2, 5]	depth=10, split=2, trees=100	0.797	0.734	0.334

Features used: rating, median_income, pct_bachelors, total_pop, median_age, distance, category (one-hot encoded)

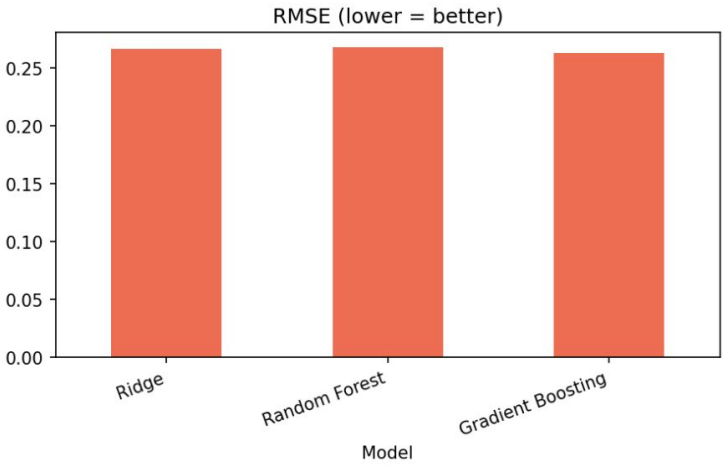
Supervised Models

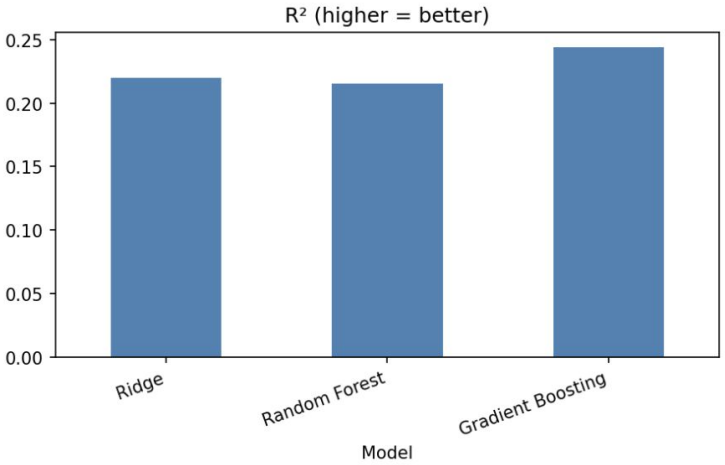
Two Tasks

- Classification - Will a business thrive?
- Regression - How successful will it be?

Models used

- Logistic Regression
- Random Forest
- Gradient Boosting





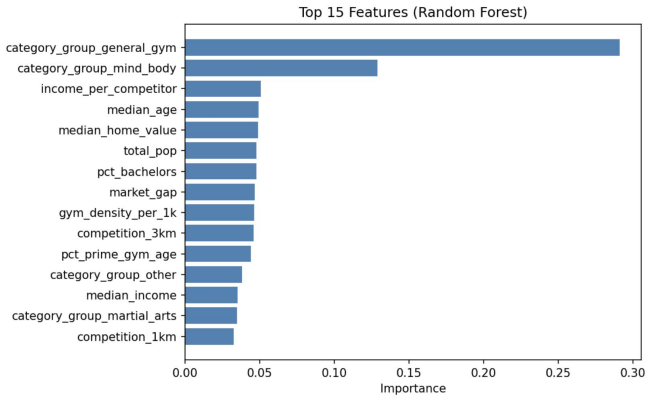
Supervised Models - Results

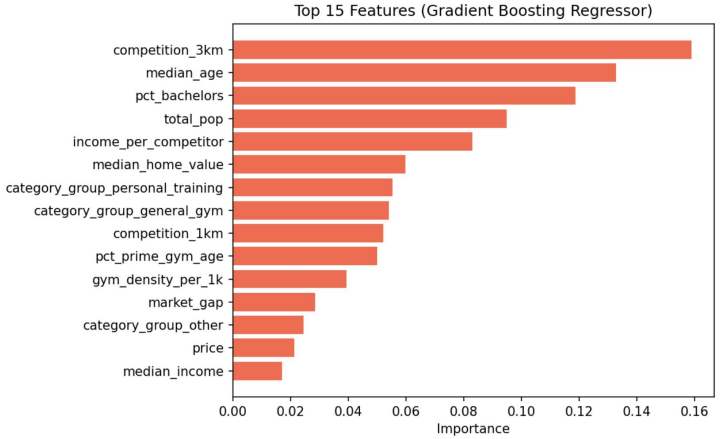
Classification

Model	Best Params	AUC	Accuracy
Logistic Regression	C=0.1	0.923	82.1%
Random Forest	depth=None, trees=200, split=2	**0.927**	**87.9%**
Gradient Boosting	lr=0.05, depth=3, trees=100	0.892	84.8%

Regression

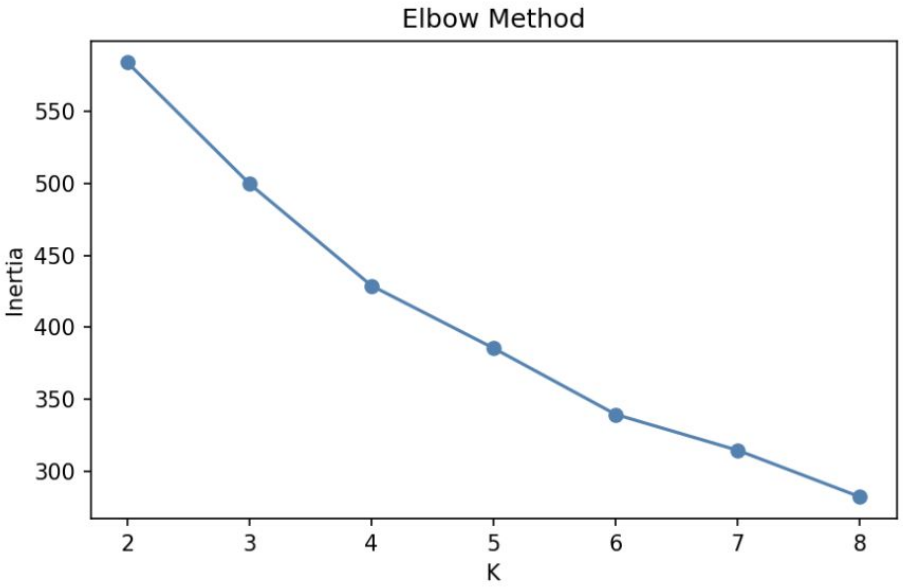
Model	Best Params	RMSE	R^2
Ridge	alpha=100	0.278	0.220
Random Forest	depth=5, trees=100	0.279	0.215
Gradient Boosting	lr=0.05, depth=3, trees=200	**0.263**	**0.244**

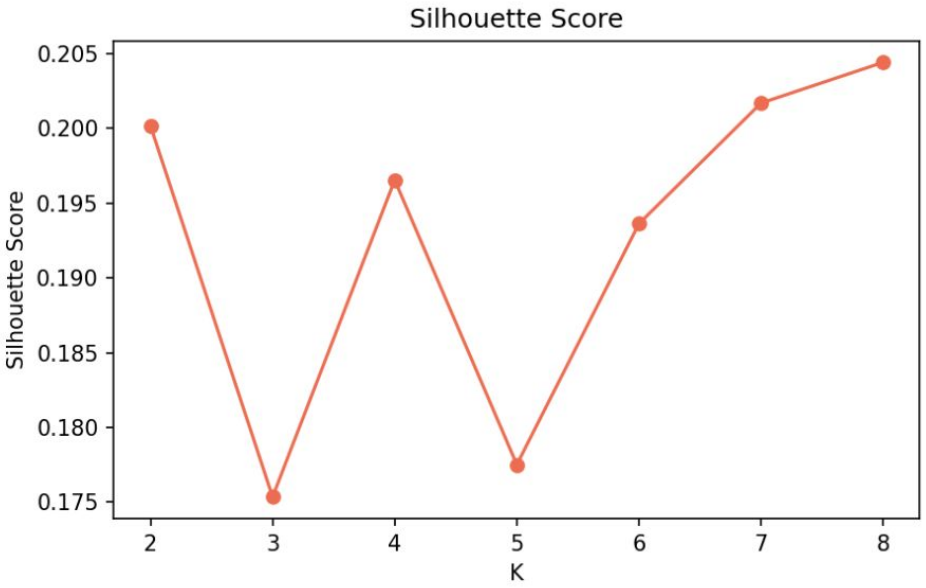


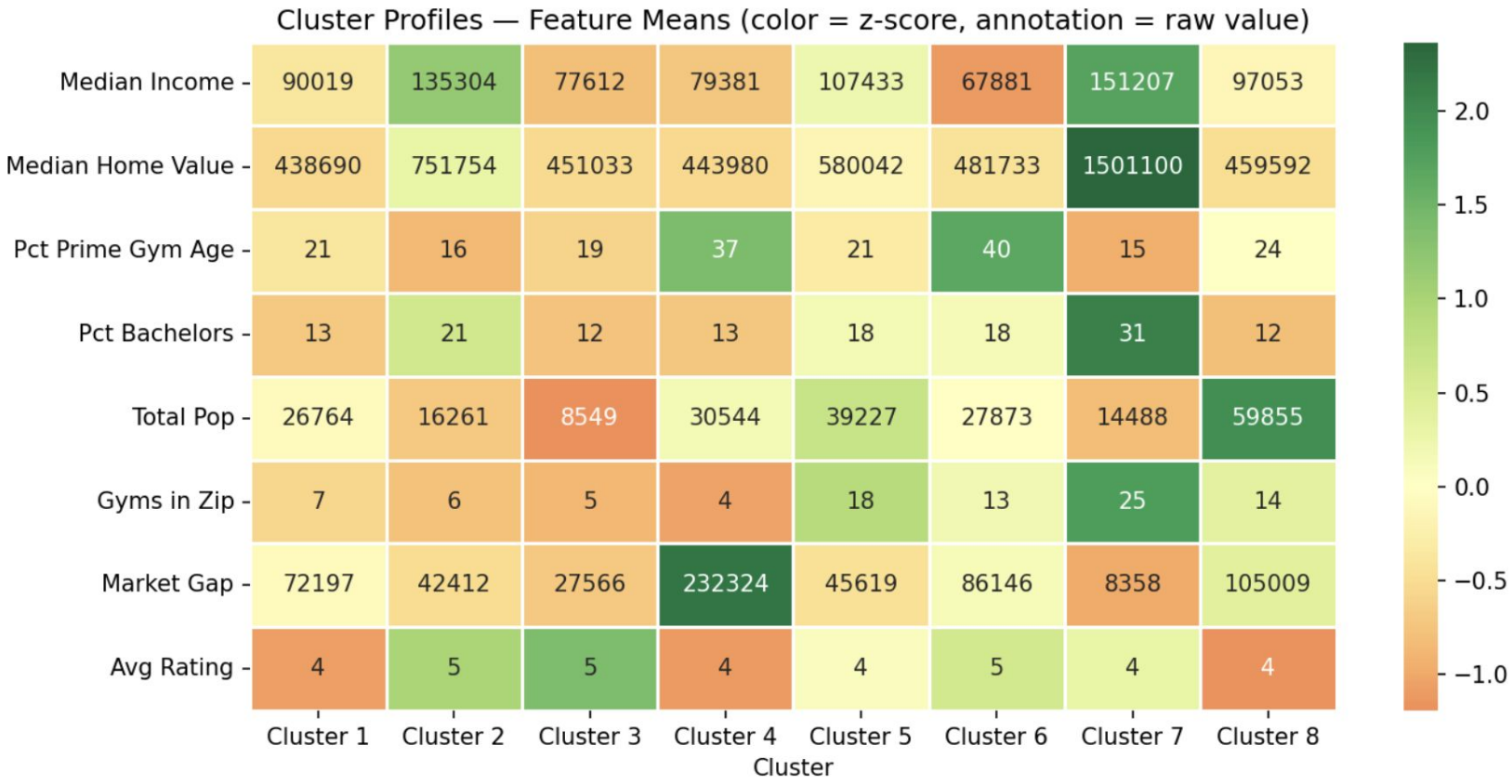


Rank	Feature	Importance
1	`rating`	0.556
2	`distance`	0.106
3	`pct_bachelors`	0.080
4	`median_income`	0.053
5	`category_gyms`	0.045
6	`total_pop`	0.039
7	`median_age`	0.033
8	`category_yoga`	0.012
9	`category_healthtrainers`	0.008
10	`category_climbing`	0.007

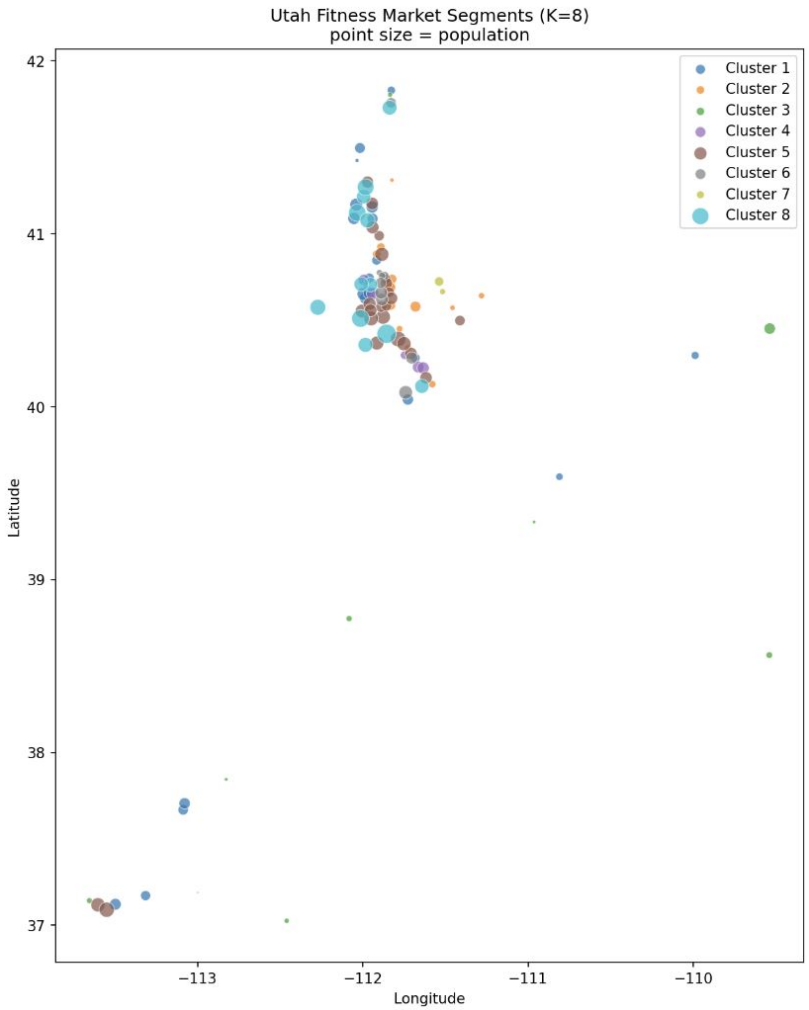
- rating dominates at 55.6% of total importance


Elbow Method
Inertia
550
500
450
400
350
300
2
3
4
5
6
7
8
K


Silhouette Score
Silhouette Score
0.205
0.200
0.195
0.190
0.185
0.180
0.175
2
3
4
5
6
7
8
K


Cluster Profiles — Feature Means (color = z-score, annotation = raw value)
Median Income
Median Home Value
Pct Prime Gym Age
Pct Bachelors
Total Pop
Gyms in Zip
Market Gap
Avg Rating
90019 135304 77612 79381 107433 67881 151207 97053
438690 751754 451033 443980 580042 481733 1501100 459592
21 16 19 37 21 40 15 24
13 21 12 13 18 18 31 12
26764 16261 8549 30544 39227 27873 14488 59855
7 6 5 4 18 13 25 14
72197 42412 27566 232324 45619 86146 8358 105009
4 5 5 4 4 5 4 4
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Cluster 6
Cluster 7
Cluster 8
Cluster
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0

Cluster	Thriving Rate	n
3	**100%**	2
2	87.5%	8
7	87.5%	8
6	77.3%	22
5	61.0%	82
1	46.7%	15
8	41.7%	24
4	**33.3%**	3



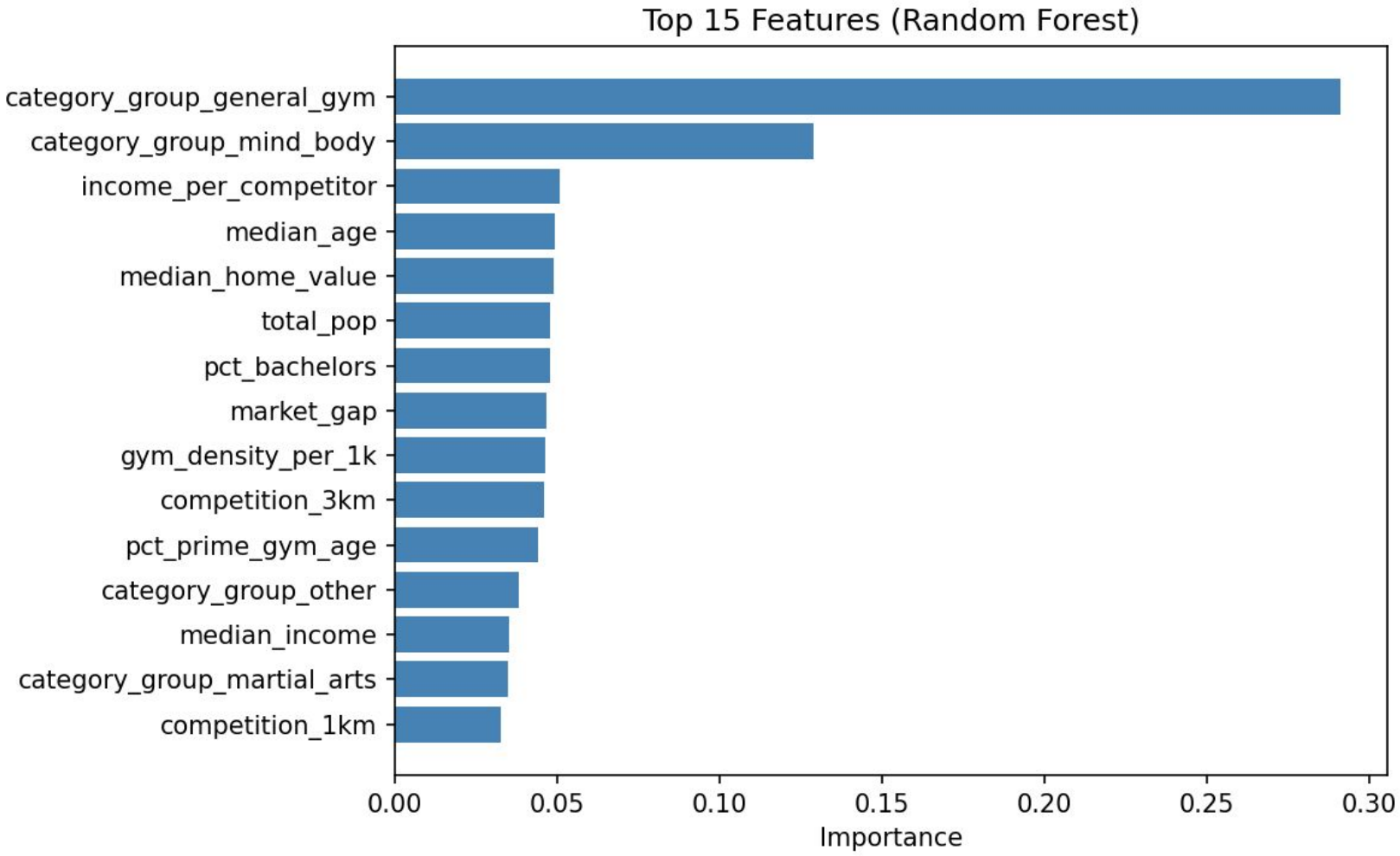
Takeaways

Fitness businesses in Utah thrive when they operate in the right *category* in the right *market context* — specifically, high-income neighborhoods where competition hasn't yet saturated demand.

- **Random Forest classifier achieves AUC = 0.927** — strong predictive signal for thriving/not thriving
- **Business Categories** have strong impact
- **Competition** is the single most useful location signal
- **K-Means reveals 8 distinct market archetypes** that explain why geographic location matters beyond just income or density alone

For a new gym owner: open a specialized gym in an affluent suburban Utah zip with few existing competitors and a large 20–44-year-old population.

Demo / Questions


Top 15 Features (Random Forest)
category_group_general_gym
category_group_mind_body
income_per_competitor
median_age
median_home_value
total_pop
pct_bachelors
market_gap
gym_density_per_1k
competition_3km
pct_prime_gym_age
category_group_other
median_income
category_group_martial_arts
competition_1km
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Importance

Additional Method Findings

Limitations and Next Steps